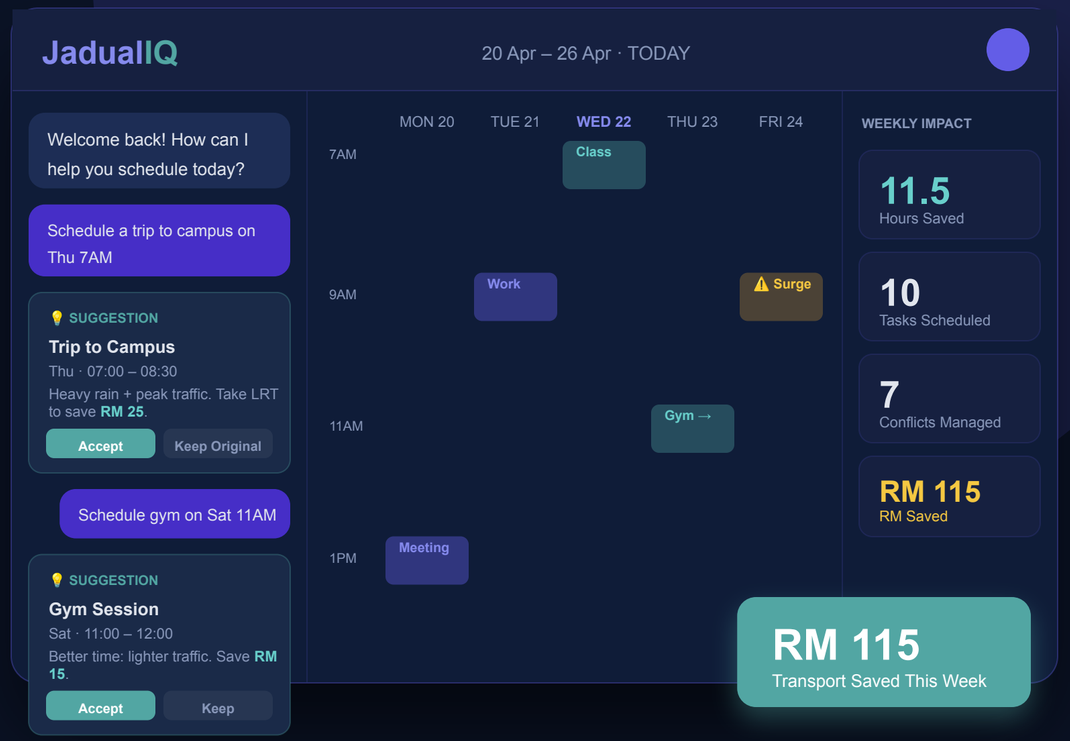
UMHACKATHON 2026 · DOMAIN 2

JadualIQ

AI Decision Copilot for **Smarter Workday Scheduling** Helping Malaysian part-time and gig workers reduce hidden transport costs.

DECISION INTELLIGENCE

SAME-DAY OPTIMIZATION



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Inefficient timing is the hidden tax on every workday

THE REALITY TODAY

📋 Part-time and gig workers **manage fragmented schedules manually** , across multiple apps and tools

🚗 Small timing decisions can create **unnecessary transport costs** that compound across the week

🌧️ Weather and traffic make those costs **unpredictable and hard to plan around**

WHAT TOOLS MISS

📅 Calendar apps **show the schedule** — they never recommend a smarter same-day alternative

🤖 AI assistants answer questions — they don't **proactively detect cost saving opportunities**

💡 No tool today **combines schedule context, weather data, and transport cost** into one actionable recommendation

Who JaduallQ is built for

The Gig Worker

Student · Part-timer · Freelancer · Any flexible worker

- Urban Malaysia
- Multiple commitments
- Ride-hailing dependent
- Budget conscious

PAIN POINTS

01 Manual schedule comparison
Checking two or three apps to decide whether a shift or appointment timing makes economic sense — every single day

02 No weather to cost awareness
No way to know when rain or peak hours will spike ride-hailing fares — until the surge is already active

03 Effort heavy decisions
Deciding whether to shift a task takes too long, so the analysis gets skipped and unnecessary cost accumulates

04 Unpredictable transport spend
End of week transport costs are a surprise, despite having a rough schedule in mind from the start

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A weekly calendar becomes a cost-aware decision assistant

Five steps. One recommendation. Always user-approved.

01

Read Schedule

Ingests the user's existing weekly calendar and identifies time flexible activity blocks

02

Check Flexibility

Determines which activities within the same day can safely be rearranged

03

Factor Context

Checks weather forecasts and travel-time data to estimate how external conditions affect cost

04

One Best Switch

GLM reasons across the tradeoffs and surfaces a single best timing recommendation

05

User Approves

The schedule stays unchanged until the user explicitly accepts the recommendation

Less manual planning
No more comparing options by hand

Lower transport cost
Optimized for same-day savings

Clearer decisions
Always explained, always optional

Four features. One economic outcome.

Weekly Schedule Optimizer

Analyzes the user's existing calendar and detects better same-day timing opportunities, without disrupting fixed commitments

Weather & Transport Cost Alert

Flags cases where weather forecasts or traffic conditions may increase expected Grab or transit cost for a planned trip

Approval Based Rescheduler

Never modifies the schedule automatically every suggested change requires an explicit user decision to accept or keep the original

Explanation Panel

Provides a plain-language rationale for every recommendation: what changed, why it saves money, and what the limitations are

Scoped to one best switch per case — to maintain clarity, trust, and decision quality.

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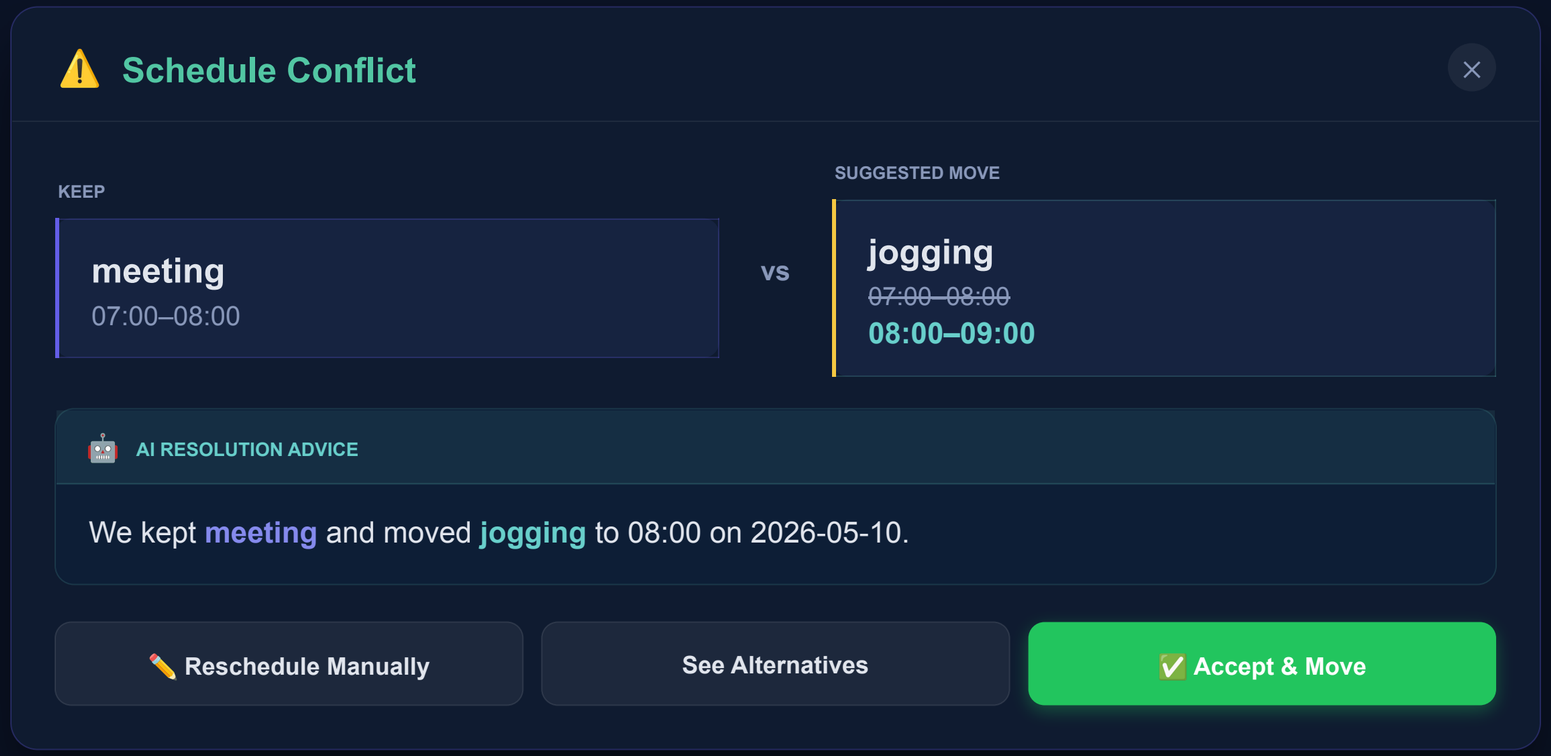
Designed for trust, not black box AI

Every recommendation shows exactly what changed, how much you save, and why before you decide anything.

- **Recommended Switch**
- Keep Original
- Weather Surge Risk
- Limited Recommendation



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One better same-day switch. Not many confusing options.

BEFORE · ORIGINAL PLAN

Aina follows her original schedule

06:45	Leaves home · Rain starts	RM 22.00
07:00	Grab booked · Surge active	+RM 6.50 surge
08:30	Arrives at campus	–
12:00	Part-time shift starts	
17:00	Shift ends · Heads home	RM 18.00

Total estimated transport
RM 40.50

→

AFTER · JADUALIQ SUGGESTS

One same-day switch · User approves

09:15	Leaves home · Rain has cleared	LRT RM 3.50
09:30	Arrives campus · Off-peak	No surge
11:00	Campus activity done ✓	–
12:00	Part-time shift starts (unchanged)	
17:00	Shift ends · LRT home	RM 3.50

Total estimated transport
RM 26.00

You saved
RM 14.50

Original schedule stays unchanged — until the user says yes

Same-day only · One switch · Full approval required

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GLM is the reasoning core not a chatbot add-on

Two distinct layers working together

🧠 GLM handles

- Contextual reasoning across schedule, weather, and travel data
- Tradeoff analysis between timing options
- Plain-language explanation generation
- Recommendation framing and confidence assessment
- Graceful handling of partial or ambiguous data

⚙️ Deterministic logic handles

- Date and time parsing
- Fixed vs flexible block enforcement
- Schedule conflict checking
- Transport cost calculations and comparisons
- Schedule reference and number validation

If GLM is removed, the system loses meaningful decision intelligence it falls back to static logic with no reasoning, no explanations, and no recommendation quality. GLM is not optional.

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Architecture designed for clear reasoning and safe validation

FRONTEND

Web App
Calendar view · Recommendation cards · Approval flow · Explanation panel

↕

BACKEND

Internal Modules
- Schedule Ingestion
- Rule / Validation
- Cost Comparison
- GLM Orchestration
- Explanation Formatter

↕

EXTERNAL APIS

Z.AI GLM
Core reasoning layer

Weather API
Forecast + rain data

Google Maps / Travel API
Traffic + travel-time context

↕

DATA LAYER

- User Schedule
- Activity Blocks
- Recommendation History
- Approval Status

> *Transport cost: API-informed estimated comparison for Grab cost using weather and Google Maps traffic data — with fallback when external data is partial.*

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Recommendation quality matters more than aggressive automation

🛡️ Decision Guardrails

- ✓ Same day changes only no multi day restructuring
- ✓ One best switch per case no option overload
- ✓ Approval required schedule never changes automatically
- ✓ Baseline comparison against original schedule always shown
- ✓ Fixed blocks are never moved regardless of cost potential

⚠️ Failure Handling

- ✓ Partial data + valid timing logic → medium confidence label
- ✓ Invalid output or incomplete data → Keep Original fallback
- ✓ Ambiguous context → Limited Recommendation label shown
- ✓ External API unavailable → graceful degradation, no crash
- ✓ Confidence and limitation notes surfaced to user always

🧪 Validation Approach

- ✓ Scenario based validation across demo cases
- ✓ Time and number consistency checks on every output
- ✓ Fallback testing under partial external data conditions
- ✓ Adversarial prompt handling and edge case coverage
- ✓ Hallucination mitigation through deterministic validation layer

Smaller scheduling decisions.
Bigger economic impact.

📈 Expected Value

Reduced estimated transport cost per week

Less time comparing scheduling options manually

Clearer, explainable same day decisions

More confidence under uncertain weather and travel conditions

🎯 Why JaduallQ Fits Domain 2

Decision intelligence not just task automation

Measurable economic outcome per session

AI-driven explanation and recommendation layer

Realistic target users with clear, validated pain points

🔗 Resources

GITHUB REPOSITORY	https://github.com/priasoloitu/UMhackathon
DEPLOYMENT	https://u-mhackathon-812r.vercel.app/

SCAN ME

JaduallQ — from fragmented schedules to smarter daily decisions